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Object Oriented Programming
Cake Order Management system

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Project Proposal

1. INTRODUCTION

The Cake Management System (CMS) is a Java-based application designed to streamline the operations of bakeries and cake shops. Using object-oriented programming principles and a command-line interface, the system provides efficient management of cake inventory, customer orders, and related data. This project demonstrates the practical application of OOP concepts including encapsulation, inheritance, and polymorphism while solving real-world business problems.

2. PROBLEM STATEMENT

Bakeries and cake shops face significant challenges in manually managing their daily operations, including:

- ✓ Tracking available cake inventory and stock levels
- ✓ Processing customer orders efficiently
- ✓ Maintaining customer records and order history
- ✓ Monitoring low-stock items to prevent shortages
- ✓ Managing order status and delivery tracking

Manual processes often lead to errors, delays, and customer dissatisfaction. There is a clear need for an automated system that can handle these operations seamlessly.

3. OBJECTIVE

3.1 Primary Objectives

- 1. Inventory Management:** Maintain a complete database of cakes with details including ID, flavor, price, and stock quantity

2. **Order Processing:** Enable users to place orders with flavor selection, quantity, and delivery specifications
3. **Customer Management:** Store and manage customer information with order history tracking
4. **Order Tracking:** Monitor order status from placement through processing to delivery
5. **Data Management:** Provide full CRUD operations for all entities (Create, Read, Update, Delete)

3.2 Technical Objectives

1. Implement Object-Oriented Programming principles
2. Develop a console-based Java application
3. Demonstrate modular programming with appropriate class structures
4. Ensure data persistence through collection frameworks

4. SCOPE

4.1 Entities Managed

1. **Cake** - ID, Flavor, Price, Stock Quantity
2. **Order** - Order ID, Customer Details, Cake Details, Status, Delivery Info
3. **Customer** - ID, Name, Contact Details, Order History
4. **Inventory Item** - Item ID, Name, Stock, Reorder Threshold

4.2 Core Operations

- ✓ Add new records
- ✓ View existing records by ID
- ✓ Update record details
- ✓ Delete records
- ✓ Low-stock alerts
- ✓ Order status tracking

5. SYSTEM ARCHITECTURE

5.1 Class Structure

Class	Purpose
CakeManagementSystem	Main entry point and user interface
Cake	Cake entity with properties and methods
Order	Order entity with status and delivery info
Customer	Customer entity with order history
InventoryItem	Inventory management with reorder alerts

5.2 Technology Stack

- ✓ **Language:** Java
- ✓ **Interface:** Command-Line Interface (CLI)
- ✓ **Data Storage:** Collections (ArrayList)
- ✓ **Paradigm:** Object-Oriented Programming

6. HOW CONDITIONAL STATEMENTS ARE USED

6.1 User Features

1. View available cakes and inventory
2. Place orders with desired flavors and quantities
3. Register as a new customer
4. Track order status

6.2 Admin Features

1. Add new cakes to inventory
2. View all records (Cakes, Orders, Customers, Inventory)
3. Update existing records
4. Delete records
5. Monitor low-stock items
6. Update inventory quantities

7. IMPLEMENTATION PLAN

Phase 1: Setup & Classes

Create class structures for all entities
Define attributes and methods
Implement constructors and getters/setters

Phase 2: Core Functionality

Implement CRUD operations for each entity
Develop order processing logic
Add inventory management features

Phase 3: User Interface

Create main menu system
Build command-line interaction
Implement input validation

Phase 4: Testing

Unit testing for each class
Integration testing
User acceptance testing

8. SAMPLE TEST CASES

Test Case	Input	Expected Output
Add New Cake	Flavor, Price, Quantity	"Cake added successfully"
Place Order	Flavor, Quantity, Customer ID	"Order confirmed"
View Order	Order ID	Display order details
Update Inventory	Item ID, New Quantity	"Inventory updated"

Test Case	Input	Expected Output
Delete Customer	Customer ID	"Customer removed"
Low Stock Alert	Automatically generated	Display items below threshold

9. LIMITATIONS

1. Console-based interface (no GUI)
2. No permanent database (data lost on exit)
3. Single-user system
4. Limited to command-line interaction
5. No automated email/SMS notifications

10. FUTURE IMPROVEMENTS

1. Database integration (MySQL/PostgreSQL)
2. Graphical User Interface (JavaFX/Swing)
3. Email/SMS notifications for orders
4. Sales reports and analytics
5. Web-based interface
6. Mobile application
7. Payment gateway integration
8. Real-time order tracking

11. CONCLUSION

The Cake Management System provides a comprehensive solution for bakeries and cake shops to manage their daily operations effectively. By automating inventory tracking, order processing, and customer management, the system reduces manual errors and improves efficiency. The object-oriented architecture ensures flexibility and maintainability, making it suitable for businesses of various sizes.